

Vasavi College of Engineering

INDUSTRIAL VISIT REPORT – IRISSET, SECUNDERABAD

(Open Elective: Signal Engineering)

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1. Introduction

On 4 November 2025, students from Vasavi College of Engineering visited the Indian Railways Institute of Signal Engineering and Telecommunications (IRISSET), Secunderabad, as part of the Signal Engineering open elective. The purpose of this visit was to bridge the gap between theoretical knowledge and practical applications in railway signalling and telecommunication systems. The visit provided exposure to advanced technologies used in the Indian Railways network, highlighting how safety, efficiency, and reliability are ensured in daily train operations.

2. About IRISSET

The Indian Railways Institute of Signal Engineering and Telecommunications (IRISSET) was established in 1957 under the Ministry of Railways, Government of India. It serves as the central training institute for engineers and staff working in the domain of signalling and telecommunications. The institute is located in Secunderabad and offers state-of-the-art laboratories, simulation facilities, and real-time setups to train personnel in both conventional and modern signalling systems. IRISSET provides training to railway officials not only from India but also from several other countries, making it a hub for knowledge exchange and technical excellence.



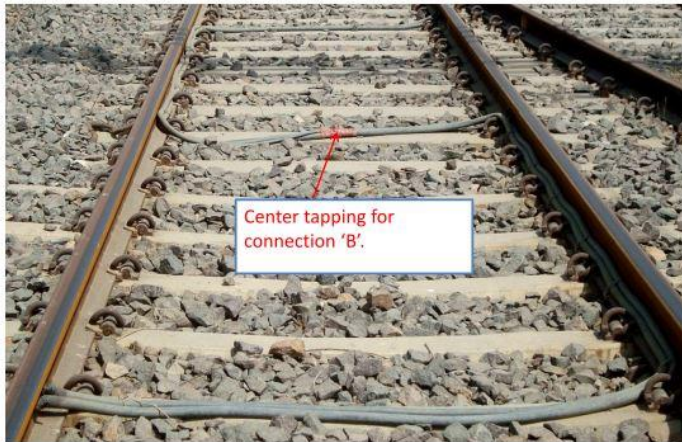
3. Objectives of the Visit

- To understand the functioning of railway signalling and communication systems in real-life scenarios.
- To study different types of bonds, circuits, and signalling equipment used in railway safety.
- To observe tokenless signalling systems and block operation in modern railways.
- To gain insights into power supply systems, IPS setups, and their redundancy.
- To comprehend the working of data loggers and their importance in monitoring and maintenance.

4. Detailed Observations

Bonds

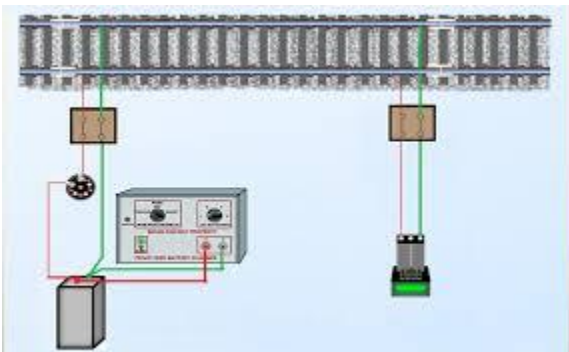
The visit began with a detailed session on various types of bonds used in railway tracks. Bonds are essential for maintaining electrical continuity throughout the rail. Students learned about structural bonds, which provide mechanical stability, and electrical bonds like longitudinal, transversal, and cross bonds, which ensure the safe return path of current. Proper bonding prevents signal failure due to voltage drop and ensures the correct operation of track circuits.



ELECTRICAL JOINT, TYPE: S BOND

Track Circuit

Track circuits are one of the oldest and most reliable train detection systems used in railways. The track circuit operates based on the principle of electrical continuity. It comprises components such as TLJB (Track Lead Junction Box), batteries, relays, and insulated rail joints (IRJs). When the track is clear, current flows through the circuit, energizing the relay and displaying a 'clear' aspect on the signal panel. When a train occupies the section, its wheels short-circuit the rails, de-energizing the relay, and thus the section is marked 'occupied'. This system is crucial for train detection and automatic signal control.



Axle Counters

The axle counter is a modern electronic device that detects train movement by counting the number of axles entering and leaving a specific track section. Each section is defined by Detection Points (DP) and Track Sections (TS). When the number of axles entering matches the number leaving, the section is considered clear. Axle counters are immune to weather effects that impact track circuits and require minimal maintenance, making them ideal for long track sections or tunnels.



Tokenless Signalling System

In earlier railway systems, tokens were used to ensure only one train occupied a block section at a time. Modern tokenless systems eliminate physical tokens by using electronic block instruments and coded bell communication between stations. During the visit, students learned about the operation of block instruments and how bell codes represent various train movements. The tokenless system enhances efficiency and minimizes human errors, improving the overall safety of railway operations.



Lock and Block System

The Lock and Block system is an advanced safety feature that interlocks signals and points to prevent conflicting train movements. When a route is set and locked for a train, other conflicting routes are automatically blocked until the section is cleared. This ensures

complete control over train movements and prevents accidents caused by human or mechanical error.

Basic Signalling System

Students also observed the basic components of railway signalling, which include signals, switches, level crossing gates, and train detection systems. Each signal aspect—red, yellow, and green—indicates whether the driver must stop, proceed cautiously, or move freely. The interlocking of signals with track devices and switches ensures that conflicting routes cannot be set simultaneously.

Power Supply System (IPS)

The Integrated Power Supply (IPS) system provides uninterrupted power to railway signalling and telecommunication equipment. It includes units like Auto Change Over (ACO), FRBW (Float Rectifier Battery Charger), ACDB (AC Distribution Board), and DCDB (DC Distribution Board). The IPS ensures redundancy by switching to backup power automatically in case of main supply failure, maintaining the reliability of signalling systems.



Data Logger

The data logger is a computerized system that records all signalling activities for analysis and maintenance. It continuously logs events such as signal changes, relay operations, and fault occurrences. By analyzing this data, railway engineers can perform predictive maintenance, diagnose issues quickly, and ensure operational safety. Data loggers act as a black box for railway signalling systems.

5. Key Learnings and Outcomes

- Understood the importance of signalling systems in ensuring railway safety and efficiency.
- Observed the integration of electrical and electronic systems in railway operations.
- Gained practical exposure to interlocking and train detection systems.
- Learned about redundancy in power supply and the role of IPS in maintaining system reliability.

- Understood how data loggers contribute to predictive maintenance and safety monitoring.

6. Conclusion

The industrial visit to IRISSET provided an enriching experience that bridged theoretical learning with practical implementation. Students gained valuable insights into how railway signalling systems operate in real-world conditions, ensuring safety, reliability, and efficiency. The knowledge acquired from this visit will help in understanding how computer science, electronics, and communication technologies integrate into complex industrial systems like Indian Railways.

7. Acknowledgment

We extend our heartfelt gratitude to the faculty members of Vasavi College of Engineering for organizing the industrial visit and to the authorities and staff of IRISSET, Secunderabad, for their cooperation and valuable demonstrations. Their guidance provided us with practical insights into railway signalling and telecommunication systems, making this visit a highly beneficial learning experience.